



Electrocution

Assessment

Pediatric Pearls:

- Use pediatric dosing of medications or electrical therapy for a pediatric patient < 37 kg and as defined by HandTevy.
- Fluids and Medication titrated to maintain a SBP >70 + (age in years x 2) mmHg
- Hypotension: (SBP < 70+ 2x Age in years)

Signs & Symptoms:

- Burns, pain, swelling
- Cardiac arrhythmias
- Cardiac Arrest
- Loss of consciousness
- Chest Pain
- Respiratory distress
- Altered mental status

Differential:

- Thermal burns
- Arrhythmia
- Chemical burns
- Ocular injury
- Respiratory arrest
- Seizures
- Syncope
- Status epilepticus
- Cardiac Arrest

Clinical Management Options

P	P	P	P	P	P	• CPR Procedure
L	L	L	L	L	L	• Evaluate for entry and exit wounds
1	2	3	4	5	6	• Elevate head of bed 30° • Declare Trauma Alert / Activation if appropriate for patient condition • Burn Clinical Guideline
						• 4 lead and 12 lead ECG placement and acquisition • ETCO₂ assessment • If airway burn: Nebulized Epinephrine for Respiratory Distress • Pain Management Guideline
						• Vascular access
						• Fluid resuscitation as indicated
						• Synchronized Cardioversion Procedure • Defibrillation • Advanced airway management • Continuous ECG monitoring

Consult Online Medical Control As Needed

Pearls:

- Electrical burns are critical burns.
- Patients with electrical injuries should be evaluated as trauma and cardiac patients receiving an ECG and cardiac monitoring.
- Prolonged monitoring is warranted in the presence of chest pain, abnormal ECG, known transthoracic path of electrical injury, loss of consciousness, and known cardiac history.
- Internal electrical injury may be significantly more severe than external thermal burns.
- AC (found in homes) is more likely to cause cardiac complications.
- DC (cars, power lines) is more likely to immobilize a patient and cause more serious burns.
- Voltage >1,000 (industrial/power lines) must go to a burn center.
- Children are likely to have oral burns from biting batteries or power cords
- Cardiac arrest due to electrocution tends to have better outcomes than other forms of arrest, even after prolonged resuscitation efforts